

Lab Assignment-4

Course: Probability and Statistics

Course code: BMAT202P

Date: 04-03-2026

Answer all the questions using R markdown and upload in VTOP.

The following data come from a hypothetical survey of 920 people (Men, Women) that ask for their preference of one of the three ice cream flavors (Chocolate, Vanilla, Strawberry). Is there any association between gender and preference for ice cream flavor?

Gender\flavor	Chocolate	Vanilla	Strawberry
Men	100	120	60
Women	350	320	150

- 1.
- 2.

As a part of quality improvement project focused on a delivery of mail at a department office within a large company, data were gathered on the number of different addresses that had to be changed so that the mail could be redirected to the correct mail stop. Table shows the frequency distribution. Fit binomial distribution and test goodness of fit

x	0	1	2	3	4
fx	5	20	45	20	10

The number of Addresses Needing Change

3. A study was performed to test whether cars get better mileage on premium gas than on regular gas. Each of 10 cars was first filled with either regular or premium gas, decided by a coin toss, and mileage for that tank was recorded. The mileage was recorded again for the same car using the other kind of gasoline. We use a paired t – test to determine whether cars get significant better mileage with premium gas.

Regular	16	20	21	22	23	22	27	25	27	28
Premium	19	22	24	24	25	25	26	26	28	

4. The Scores of 10 candidates prior and after training are given below

Prior 84 48 36 37 54 69 83 96 90 65

After 90 58 56 49 62 81 84 86 84 75

Test the training is Effective or Not?

5. An IQ test was administered to 5 persons before and after they were trained. The results are given below:

Candidates I II III IV V

IQ before Training 110 120 123 132 125

IQ After Training 120 118 125 136 121

Test whether there is any change in IQ after the training Programme.